ChronoFlow: an electricity-free, volumetric, clockwork-governed syringe infusion driver for neonatal care in low-resource settings

Author list to be completed[1]

[1]Affiliation to be completed

Design-and-simulation study — October 3, 2026

Abstract

Hospitals without reliable electricity deliver intravenous fluids and drugs to newborns by gravity drip or by hand. The only non-electric infusion devices available meter flow by pushing fluid through a restriction, so the delivered rate depends on fluid viscosity (temperature, drug) and on downstream pressure; low-cost electric pumps still need batteries and cost hundreds of dollars. We present ChronoFlow, a wind-up syringe driver that meters *volume* rather than pressure: a constant-force spring pushes the plunger and a rack-and-pinion drives a 3D-printed step-up gear train into a clockwork escapement whose balance wheel sets the rate. Two design problems were identified and solved by simulation. First, the mechanical power available to the escapement varies 18× across clinical rates and loads (15–275 µW); a conventional lever escapement, whose impulse scales with train torque, over-banks over most of this range. Adapting the gravity-escapement principle of tower clocks with springs—a constant-force *spring-arm* escapement—delivers a fixed impulse energy per beat and holds the balance amplitude constant over the full range with a period error below 0.06 %. Second, a cross-spring flexure pivot was shown by a beam-column finite-element model to change stiffness by 17 % under the balance's own weight; it was replaced by a taut-band torsion suspension made from 0.04 mm feeler-gauge steel ribbon, giving a 1.20 Hz balance with ±0.16 % orientation sensitivity and no pivot friction. Colour-coded gear cassettes provide 1, 2, 5 and 10 mL/h on a 20 mL syringe (0.4–19.5 mL/h across 5–60 mL syringes) at 0.12–1.16 µL per beat. A 20 000-sample Monte-Carlo error budget predicts that 95 % of units deliver within ±1.14 % of the set rate, compared with +45 % at 37 °C and −33 % at 100 mmHg back-pressure for flow-restriction pumps. A calibrated trip spring stops the oscillator and raises a flag at 300 mmHg occlusion and a syringe-latch interlock prevents free-flow boluses. The complete 66-part assembly was modelled parametrically, checked for interferences and exported for printing; the bill of materials is $38. No physical prototype has yet been built; a bench protocol following IEC 60601-2-24 is provided.

1 Introduction

Neonates require maintenance fluids at 60–100 mL/(kg d) and drugs such as antibiotics, dextrose and inotropes at 0.5–10 mL/h; a 10 % rate error or an uncontrolled bolus of a few millilitres is clinically significant. Electronic syringe pumps meet this need where mains power, batteries and service infrastructure exist. Where they do not, three alternatives have been described. Low-cost electromechanical pumps such as AutoSyP combine a constant-force spring with a motor and battery and reach ±4 % accuracy at a cost of goods of about $500 [1]. Non-electric flow-restriction devices (elastomeric pumps, the Springfusor, and an adjustable air-over-oil design [2]) obey $Q \propto \Delta P/\mu$ and therefore drift with fluid viscosity, temperature and back-pressure. Two clockwork devices have been reported: a gravity-driven driver regulated by a pendulum metronome, which must sit level and still [3], and a kitchen-timer movement adapted for microfluidics with one fixed rate and about one hour of running time [4].

The gap is a non-electric device that meters *volume*, works in any orientation, offers selectable clinical rates, runs for a full syringe without re-winding and has mechanical occlusion detection

and free-flow protection. ChronoFlow addresses it by returning to two solutions from precision mechanics: the constant-force escapement, introduced by Denison for tower clocks whose train torque varied with wind and ice on the hands [5], and the taut-band suspension of galvanometers, which removes pivot friction and most positional error.

2 Design

2.1 Architecture and metering principle

Figure 1a–b shows the machine. Two constant-force springs (30 N total) pull a carriage toward the syringe nozzle. The plunger load—syringe friction of 2–5 N plus pressure times bore area—absorbs part of this force; the excess back-drives a module-1 rack and 12-tooth pinion through a $5 \times 5 \times 5 \times 2.5 = 312.5$ step-up train and a plug-in rate cassette to a 15-tooth escape wheel. The escapement releases one half-tooth per beat of the balance, so the plunger advance per beat is fixed by geometry:

$$\Delta x = \frac{\pi m z_0}{2 Z_e \, N_{\text{fixed}} \, r_{\text{c}}} = \frac{\pi \cdot 1 \cdot 12}{2 \cdot 15 \cdot 312.5 \cdot r_{\text{c}}} \ \text{mm}, \tag{1}$$

which is 4.0 µm (1.16 µL on a BD 20 mL syringe) for the unity cassette. Because the metered quantity is displacement, viscosity, temperature and downstream pressure do not enter the rate until the occlusion trip point (Fig. 1c–d). The spring force is constant over the 95 mm travel, so runtime is set only by syringe volume (2–20 h per 20 mL fill) and the device never needs re-winding mid-infusion. A one-way click on the rack-pinion shaft lets the carriage be pulled back freely, so loading a syringe *is* the winding action (2.85 J stored, about eight times the 0.37 J the escapement dissipates in 20 h).

2.2 Constant-force spring-arm escapement

Power delivered to the escapement is $P = F_{\text{excess}} \, v$. With F_{excess} between 15.5 N (at the occlusion trip point) and 28.5 N (free-running, 5 mL syringe) and v spanning the 1–10 mL/h cassettes, P ranges from 15 µW to 275 µW; after a train efficiency of 0.58 (rack 0.95, seven printed meshes at 0.92) the escape-wheel torque spans 17–315 µN m, an 18.4× range. In a lever escapement the impulse energy delivered to the balance is proportional to this torque, so the balance amplitude scales as $\sqrt{T_e}$; our dynamic model (Section 3) shows it exceeding the 1.2 rad banking limit over most of the operating range (Fig. 2b–c). Stalling at the low end or over-banking at the high end are both failures of rate.

The spring-arm escapement (Fig. 2a) separates the two energy flows. Each escape-wheel tooth *cocks* one of two impulse arms against a printed flexure until a latch engages (phases 1–2). The train is then locked and any excess torque simply sits on the lock. When the balance passes, its impulse pin trips the latch (phase 3) and the arm delivers a fixed energy $E_{\text{imp}} = 1.2$ µJ over a 15° arc; the wheel then advances one half-pitch, re-cocks the opposite arm and relocks, the surplus train energy being dissipated in the locking impact. Train torque only has to exceed the re-cock minimum $T_{\min} = E_{\text{imp}}/(\eta_{\text{lift}} \, \pi/Z_e) = 9.5$ µN m. The simulation confirms constant amplitude (0.59 rad at $Q = 150$) and a period error of 0.06 % from the slightly asymmetric impulse window across the whole torque range (Fig. 2c–d). Because the balance's air damping cannot be known before a prototype exists, Q was swept from 40 to 800: amplitude settles between 0.31 and 0.99 rad and the period error between 0.45 and 0.01 % (Fig. 2e), a fixed offset that trim calibration absorbs.

2.3 Taut-band balance

The oscillator (Fig. 3a) is a printed PETG ring (62 mm outer diameter, 7.4 g, $I = 5.2 \times 10^{-6}$ kg m^2) clamped at the midpoint of a 0.04 mm × 2 mm spring-steel ribbon cut from a 0.0015 in feeler-gauge blade, held at both ends and pre-tensioned to $T_0 = 1$ N by a steel coil spring. Each 25 mm half-band

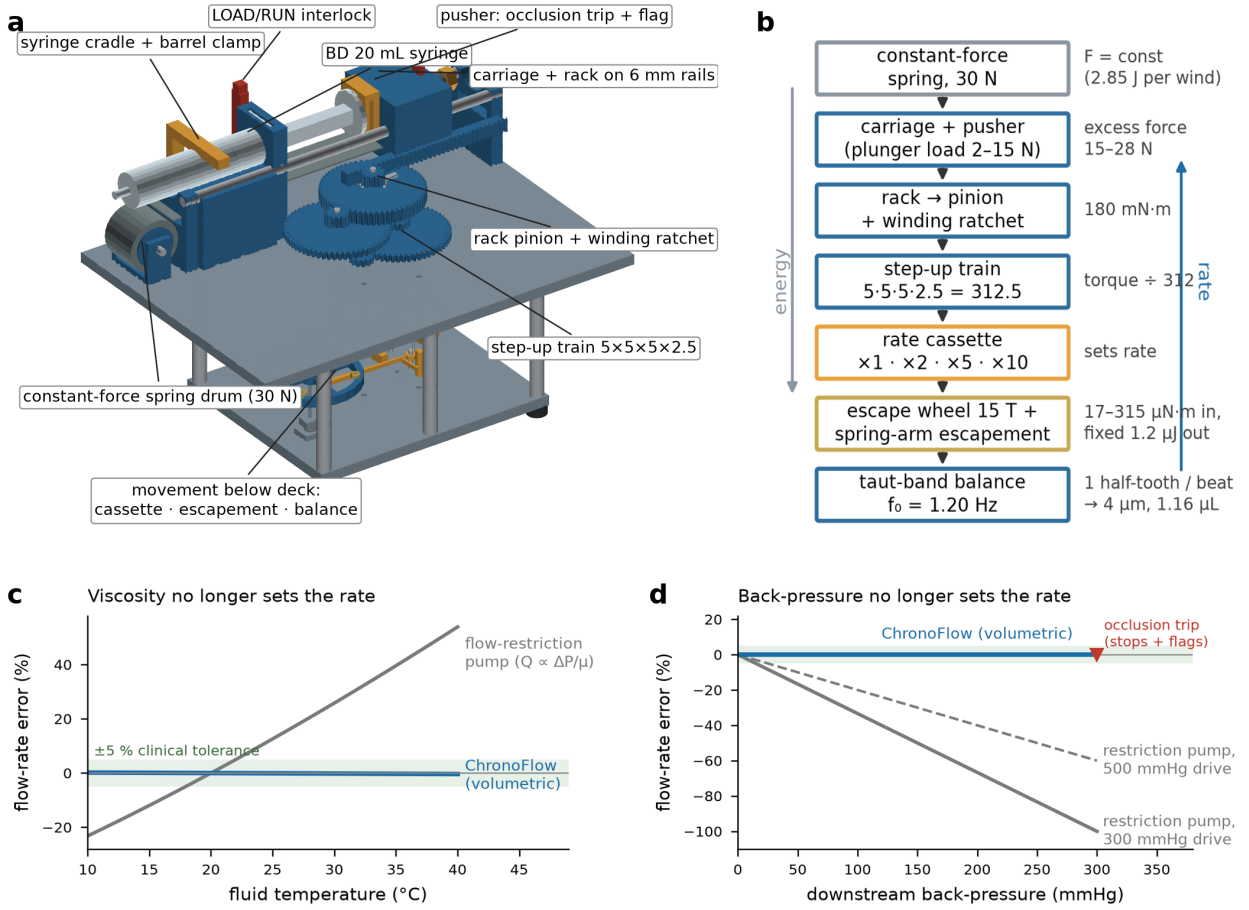


Figure 1: A spring-driven, clockwork-governed syringe driver meters volume, so the delivered rate is independent of fluid temperature and back-pressure. (a) Labelled render of the parametric CAD assembly with a BD 20 mL syringe at 85 % fill and the 10 mL/h cassette. (b) Energy and metering chain; the left arrow follows energy from the spring to the balance, the right arrow follows the rate constraint back from the balance to the plunger. (c) Steady-state flow error versus fluid temperature for a flow-restriction pump calibrated at 20 °C (water viscosity from the Vogel equation) and for ChronoFlow (oscillator temperature coefficient $-1.8 \times 10^{-4}\,\mathrm{K}^{-1}$). (d) Flow error versus downstream back-pressure; ChronoFlow stops and flags at the 300 mmHg trip point. Green band: ±5 % clinical tolerance.

is a torsion spring of stiffness

$$k_{\mathrm{half}} = \frac{GJ}{L_h} + \frac{T\,w^2}{12\,L_h}, \qquad J = \tfrac{1}{3} w t^3 \left(1 - 0.63\,\tfrac{t}{w}\right), \tag{2}$$

giving $k = 2.96 \times 10^{-4}\,\mathrm{N\,m/rad}$ and $f_0 = 1.20\,\mathrm{Hz}$. Four M3 brass nuts in radial slots (23–29 mm) regulate the frequency by −2.2 to +3.2 % (Fig. 3b), exactly as a clock is regulated; the nominal set point is 26.6 mm.

Gravity enters only through the tension term: with the band vertical, the fixed-clamp half-band carries $T_0 \pm mg$ while the tensioner fixes the other at T_0. Because the tension term is 9 % of k and $mg \ll T_0$, the frequency shifts by only ±0.16 % between the upright and inverted positions (Fig. 3c). An alternative two-leaf cross-spring flexure pivot, analysed with an Euler–Bernoulli beam-column finite-element model including geometric stiffness, changed rotational stiffness by 17 % under the same balance weight (the governing scale is $EI/L^2 \approx 0.1\,\mathrm{N}$, not the Euler load) and was rejected. Peak shear stress at the 1.2 rad banking limit is 154 MPa against a conservative fully-reversed fatigue limit of 350 MPa (Fig. 3d), and the temperature coefficient, combining the shear-modulus drift of steel with the thermal expansion of the ring, is $-1.8 \times 10^{-4}\,\mathrm{K}^{-1}$ (Fig. 3e).

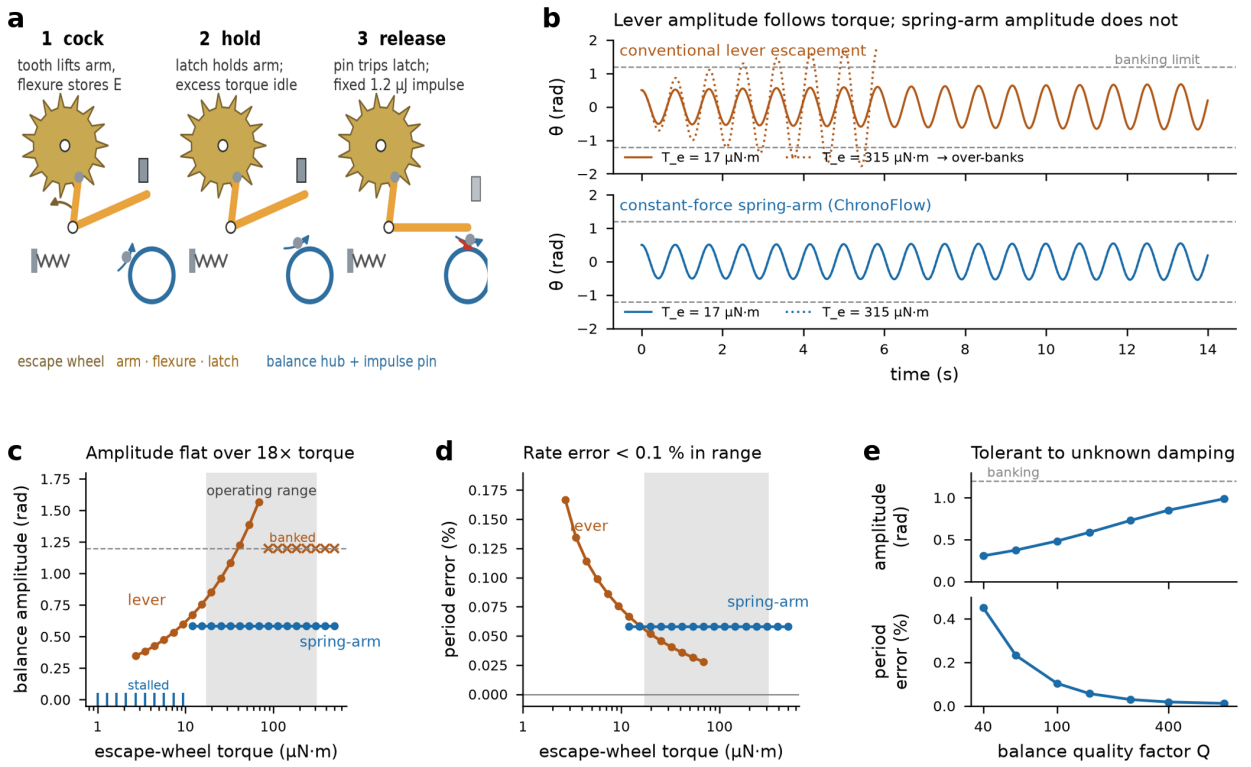


Figure 2: A constant-force spring-arm escapement keeps balance amplitude and rate independent of the 18× variation in train torque that defeats a conventional lever escapement. (a) Working principle: a wheel tooth cocks a spring-loaded impulse arm (1), a latch holds it while the wheel is locked (2), the balance pin trips the latch and receives a fixed impulse (3). (b) Simulated balance angle $\theta(t)$ for both escapements at the minimum and maximum operating torques (17 and 315 µN m; RK4, $Q = 150$); dashed lines mark the banking limit. (c) Steady amplitude and (d) period error relative to the free oscillator versus escape-wheel torque; shaded band: operating range. Ticks at zero mark torques below the spring-arm re-cock minimum. (e) Spring-arm amplitude and period error versus the balance quality factor Q.

2.4 Rate cassettes, safety functions and structure

A plug-in cassette carries two gear pairs with constant tooth sums ($z_1 + z_2 = z_3 + z_4 = 66$, 33 mm centres): (33, 33, 33, 33), (44, 22, 33, 33), (33, 33, 55, 11) and (44, 22, 55, 11) give speed-up ratios of 1, 2, 5 and 10 and hence 10, 5, 2 and 1 mL/h on a 20 mL BD syringe; other syringe sizes scale with bore area (Fig. 4a). The cassette is a visible, colour-coded object, so the set rate can be confirmed at a glance and cannot drift. The escape-wheel torque exceeds the re-cock minimum by at least 1.79× at every rate down to the occlusion trip point (Fig. 4b), and the per-beat volume is 0.12–1.16 µL (Fig. 4c).

Three safety functions are mechanical. (i) The pusher head rides on a stem with a knob-adjustable compression spring (8–20 N); when the plunger reaction exceeds the preload the stem retracts 3 mm, exposes a red flag and lifts a trip rail that blocks the balance, so ticking stops and pressure cannot rise further. The default 14.5 N corresponds to 300 mmHg on a 20 mL syringe; settings for 5, 10, 30 and 60 mL syringes are 6.1, 8.6, 18.3 and 26.9 N. (ii) The LOAD/RUN lever opens the syringe flange latch and lifts the rack-pinion click together, so the carriage can only be moved by hand when the syringe is unlatched and a fast-forwarded pusher slides the whole syringe rather than expelling fluid. (iii) The balance beats behind a window at 1.2 Hz; silence means stopped.

The most heavily loaded teeth are the rack and 12-tooth pinion: Lewis bending stress at 30 N on an 8 mm face is 15.3 MPa against about 50 MPa PETG yield (safety factor 3.3). Torque falls 5× per stage (180, 33, 6.1, 1.1 mN m), so the remaining printed gears are lightly loaded; shafts are 3 mm steel pins in printed bores.

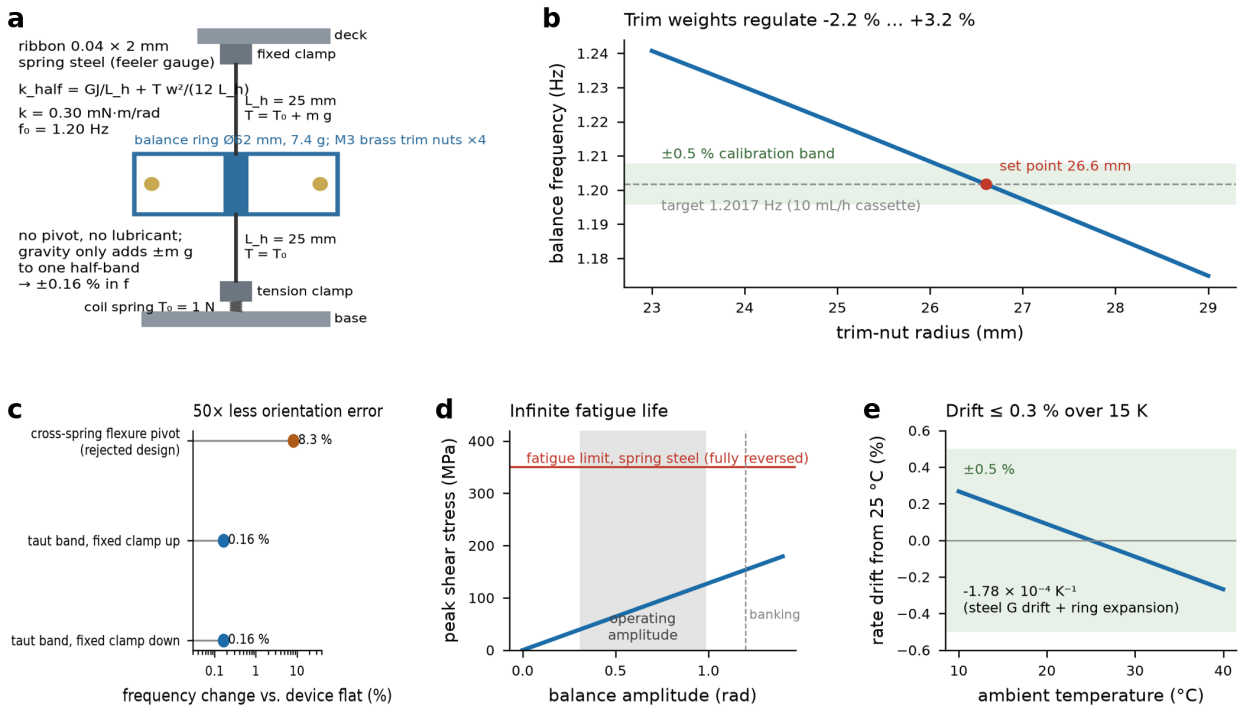


Figure 3: A taut-band suspension gives a friction-free 1.2 Hz balance whose rate is insensitive to orientation, temperature and fatigue. (a) Construction: feeler-gauge ribbon clamped to the deck and to a spring-loaded tension clamp, balance ring clamped at mid-span. (b) Balance frequency versus radial position of the four trim nuts; the dashed line is the target for the unity cassette and the band is ±0.5 %. (c) Frequency change between the band-horizontal and band-vertical positions for the taut band, compared with the computed first-order change of a 0.05 mm cross-spring flexure pivot under the same balance weight. (d) Peak ribbon shear stress versus amplitude; shaded band: amplitudes reached for $Q = 40$–800. (e) Rate drift versus ambient temperature.

3 Methods

Parametric design. All dimensions, tooth counts and oscillator parameters live in one module (`design_params.py`); the CAD (`cad_model.py`, CadQuery 2.8) derives shaft positions from gear centre distances, phases every mesh so that a wheel tooth faces a pinion gap, and exports STEP, glTF and per-part STL. A pairwise interference check (bounding-box prefilter, OpenCascade Boolean common) over all 2 145 part pairs reports no unintended interferences; the single residual is a 0.55 mm^3 tip/root contact on the 60T/12T mesh, the known undercut of a 12-tooth 20° pinion, for which a +0.3 profile shift is recommended.

Escapement dynamics. The balance obeys $I\ddot{\theta} = -k\theta - c\dot{\theta} + \tau(\theta, \dot{\theta})$ with $c = I\omega_0/Q$, integrated by fourth-order Runge–Kutta at $\Delta t = 0.2$ ms. For the lever model the per-beat wheel energy $T_e\pi/Z_e$ is split into an impulse fraction (0.70) applied over a 30° lift arc and an unlocking fraction (0.12) opposing motion over 6°. For the spring-arm model the impulse energy is fixed at E_{imp} over 15° starting 3° before equilibrium, with a fixed 0.12 µJ latch-tripping cost; the escapement stalls if $T_e < T_{\min}$. A reversal of the balance while the pin is inside the engagement window is treated as a fault. Period is measured over steady-state zero crossings.

Oscillator. Band stiffness follows Timoshenko's thin-strip torsion with the tension term; the cross-spring pivot comparison uses a 60-element Euler–Bernoulli beam-column model per leaf (elastic plus geometric stiffness), with the balance rotating rigidly about the crossing point and gravity resolved onto the two leaves.

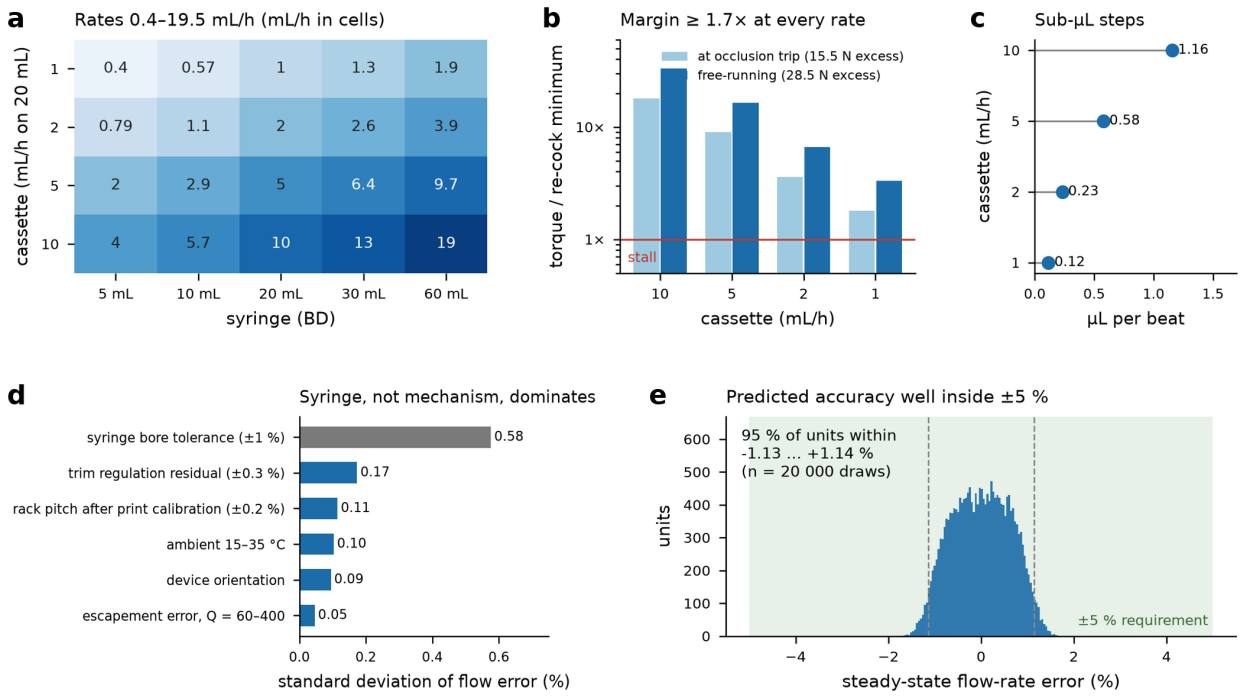


Figure 4: Four plug-in cassettes cover neonatal rates with torque margin and sub-microlitre resolution; the predicted error is dominated by the syringe, not the mechanism. (a) Flow rate for each cassette and BD syringe size. (b) Escape-wheel torque relative to the arm re-cock minimum at the occlusion trip point and free-running. (c) Volume delivered per beat. (d) Standard deviation of flow error contributed by each Monte-Carlo input. (e) Distribution of steady-state flow error over 20 000 draws; dashed lines mark the central 95 %.

Error budget. 20 000 Monte-Carlo draws combine uniform syringe bore tolerance (±1 % area), trim regulation residual (±0.3 %), rack pitch after print-scale calibration (±0.2 %), ambient temperature (15–35 °C), orientation (±0.16 %) and the escapement error for Q uniformly in 60–400 after calibration at the mean.

Comparators. Flow-restriction pumps are modelled as $Q \propto (P_{\mathrm{drive}} - P_{\mathrm{back}})/\mu(T)$ with water viscosity from the Vogel equation ($\mu = 0.02939 \exp[507.88/(T - 149.3)]$ mPa s) and drive pressures of 300 and 500 mmHg.

4 Results

Table 1 collects the predicted performance. Two results carry the design. First, the escapement study (Fig. 2) shows that the conventional lever escapement is unusable across the clinical range—its amplitude passes the banking limit at about 40 µN m, below the mid-point of the operating band—whereas the spring-arm escapement holds 0.59 rad from the re-cock minimum upward with a period error of 0.06 %. Second, the suspension study (Fig. 3) shows that the taut band reduces orientation sensitivity by a factor of about fifty relative to a cross-spring pivot of the same leaf stock, while keeping the ribbon stress at 44 % of the fatigue limit at the banking amplitude. The system-level consequences (Fig. 4) are a rate set entirely by integers (tooth counts) and one calibrated frequency, a torque margin above the re-cock threshold at every rate, and a predicted accuracy distribution whose width is set mainly by the syringe bore tolerance that every syringe pump shares.

Against flow-restriction pumps the difference is categorical rather than incremental (Fig. 1c–d): a restriction pump calibrated at 20 °C over-delivers by 45 % at body temperature and loses a third of its rate against 100 mmHg of venous or catheter back-pressure, whereas ChronoFlow drifts by less than 0.4 % over 10–40 °C and is unaffected by back-pressure until it trips.

Table 1: Predicted performance of ChronoFlow (BD 20 mL syringe unless stated) and bill of materials.

Quantity	Value
Rates (cassettes 1/2/5/10)	1, 2, 5, 10 mL/h; 0.4–19.5 mL/h across 5–60 mL syringes
Resolution	0.12–1.16 µL per beat, 2.4 beats s^{-1}
Oscillator	1.2017 Hz; regulation −2.2/+3.2 %; orientation ±0.16 %; $-1.8 \times 10^{-4}\,K^{-1}$
Escapement	amplitude 0.59 rad ($Q = 150$), period error 0.06 %; torque margin $\geq 1.79\times$
Accuracy (Monte-Carlo, $n = 20\,000$)	95 % of units within −1.13 to +1.14 %; 99th percentile |error| 1.35 %
Restriction-pump comparator	+45 % at 37 °C; −23 % at 10 °C; −33 % at 100 mmHg back-pressure
Occlusion trip	300 mmHg (knob-set per syringe size); oscillator stopped, flag raised
Energy	2.85 J stored per wind; 0.37 J consumed in 20 h
Structure	rack-pinion tooth stress 15.3 MPa (SF 3.3); band stress 154 MPa vs 350 MPa limit
Parts, cost	66 parts, ≈420 g PETG, BOM ≈ $38

The CAD model (Fig. 5) is complete at the part level: deck and bottom plates, pillars, rails, cradle with flange latch and barrel clamp, carriage with integrated rack and pusher assembly, spring drum, the four-stage train, cassette, escape wheel, two impulse arms with flexures and latch, and the balance with band, clamps and tensioner.

5 Discussion

ChronoFlow converts two precision-mechanics solutions—the constant-force escapement and the taut-band suspension—into a printable, electricity-free infusion driver whose predicted accuracy is limited by the syringe rather than by the fluid, the temperature, the back-pressure or the orientation of the device. Every safety function is mechanical and visible, and the rate is a physical object a nurse can hold and verify.

Several limitations follow from the fact that this is a design-and-simulation study. All performance figures are model predictions; the models are physics-based but unvalidated until the bench protocol below is executed. The balance quality factor in air and the efficiency of printed gear meshes are the least certain inputs; both were swept and both affect margin rather than rate. The spring-arm escapement geometry in the CAD is a representative embodiment whose latch and arm proportions will need iteration on the bench, as horological escapements invariably do. The occlusion trip must be set per syringe size, which is a training requirement, and rates between cassettes require an additional printed cassette (any tooth set with sums of 66 can be made). The 0.12–1.16 µL pulsatility is small compared with a 50 µL gravity-drip drop, but very low-rate inotrope infusions below 0.5 mL/h would be better served by a 5 mL syringe.

Validation protocol. (1) Print the balance, clamp the ribbon and measure f_0 with a 240 fps camera or light gate; set trims; measure Q from the free-decay envelope; repeat band-horizontal, clamp-up and clamp-down (acceptance ±0.3 %). (2) Drive the escape wheel through the cassette with hanging weights emulating 15 and 28 N excess; verify re-cocking at the 1 mL/h cassette and constant amplitude by video tracking; tune arm flexure thickness. (3) Gravimetric flow test per IEC 60601-2-24 on a 1 mg balance logged every 30 s for 2 h per cassette; compute trumpet curves (acceptance: mean error ±5 %, trumpet ±10 % at 2 min). (4) Repeat at 0, 100 and 200 mmHg back-pressure and with 10 % dextrose at 20 and 37 °C (expected change < 1 %). (5) Occlusion: clamp the line, record trip pressure ((300 ± 30) mmHg), time to trip and bolus on release (< 0.2 mL). (6) Orientation and vibration (expected < 0.5 %). (7) Seven-day continuous run and 200 load/unload cycles.

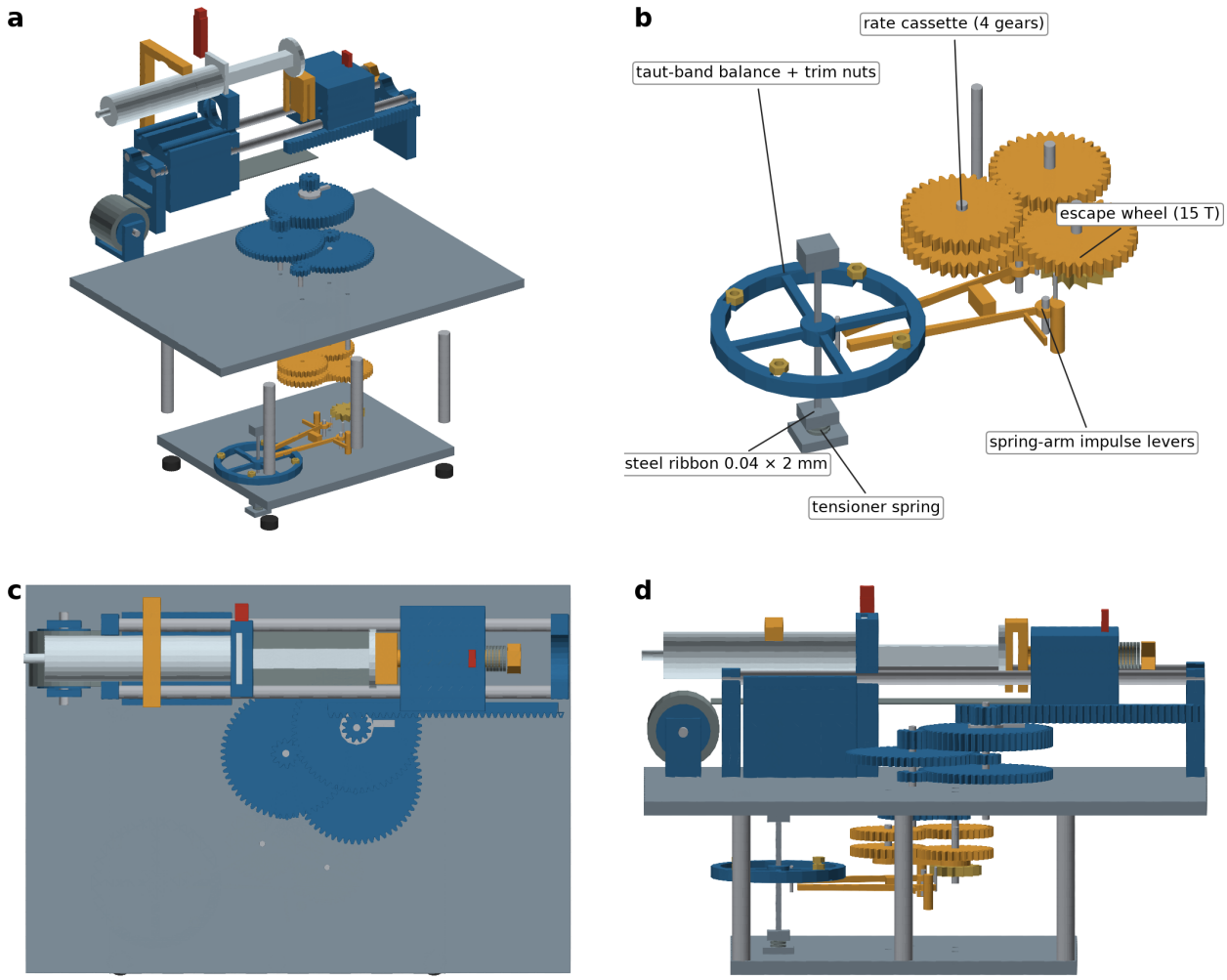


Figure 5: The complete 66-part assembly is fully modelled, interference-checked and printable. (a) Exploded view with parts separated along the vertical axis. (b) Escapement module with the deck removed. (c) Top view of the upper deck and step-up train. (d) Front elevation showing the movement below the deck.

Data and code availability

The parametric CAD (STEP, glTF, STL), the analysis code that reproduces every figure and number in this paper, the interference checker and the interactive viewer are provided in the accompanying repository.

References

[1] Juarez A, Maynard K, Skerrett E, Molyneux E, Richards-Kortum R, Dube Q, Oden ZM. AutoSyP: a low-cost, low-power syringe pump for use in low-resource settings. *Am J Trop Med Hyg* 2016;95(4):964–969. doi:10.4269/ajtmh.16-0285.

[2] Design of a novel, adjustable flow rate, reusable, electricity-free, low-cost syringe infusion pump. *ASME J Med Devices* 2017;11(4):041006.

[3] Design of a novel mechanical syringe pump for neonatal care in low-resource settings. ASME International Mechanical Engineering Congress & Exposition, 2011.

[4] Non-electric, spring-powered and clockwork syringe pumps for microfluidic devices. *Sens Actuators B* 2024; *HardwareX* 2024.

[5] Denison EB. *A Rudimentary Treatise on Clocks, Watches and Bells*. London: Weale; 1850 (gravity escapement).

[6] Timoshenko SP, Goodier JN. *Theory of Elasticity*, 3rd ed. McGraw-Hill; 1970 (torsion of thin rectangular sections).

[7] IEC 60601-2-24: Medical electrical equipment — particular requirements for the basic safety and essential performance of infusion pumps and controllers.

[8] Vogel H. Das Temperaturabhängigkeitsgesetz der Viskosität von Flüssigkeiten. *Phys Z* 1921;22:645–646.